ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A3 · LIFE CYCLES

Life cycle stages

★☆☆
Grade 4

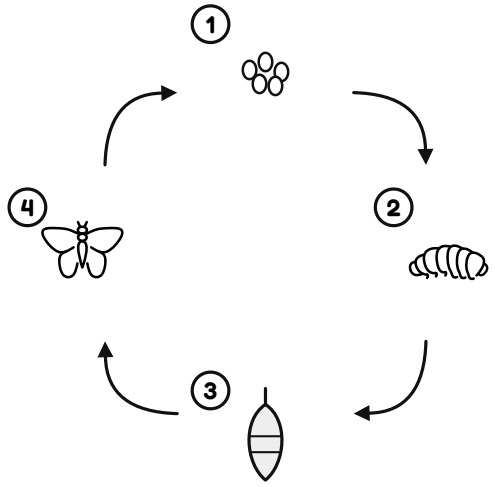
Learning goals

- Order life cycle stages.
- Use life cycle vocabulary.

1 **Number the stages of a frog's life cycle in order.** [4]

- [5] an adult frog
- [1] eggs (frogspawn) in the water
- [4] a froglet with a short tail
- [2] a tadpole with a tail and gills
- [3] a tadpole with back legs

2 **Label the butterfly life cycle.** [4]



Number	Stage
1	eggs
2	caterpillar (larva)
3	chrysalis (pupa)
4	adult butterfly

3 **Fill in the gaps.** [4]

WORD BANK

metamorphosis **inherit** **hatches** **germinates** **migrates** **melts**

A seed germinates into a seedling. A chick hatches from an egg. A big change of body shape is called metamorphosis. Young living things inherit features from their parents.

4 **Sort the features.** [6]

SORT THESE

shape of nose **eye colour** **a scar on your knee** **a haircut** **a puppy's fur colour** **the language you speak**

Inherited from parents	Not inherited
eye colour	a scar on your knee
shape of nose	the language you speak
a puppy's fur colour	a haircut

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A3 · LIFE CYCLES

Comparing life cycles

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Grade 4

Learning goals

- Compare life cycles.
- Explain variation.

1 **Give one way the life cycle of a frog is similar to a chicken's and one way it is different.** [2]

Similar: both start as eggs and grow into adults that lay eggs. Different: the frog changes shape (tadpole to frog), but the chick already looks like a small adult.

2 **Why does a butterfly lay its eggs on a leaf and not on a stone?** [2]

When the caterpillars hatch they need leaves to eat straight away.

3 **Five kittens are born to the same mother. Two are black, two are striped and one is white. Explain why they are not all the same.** [2]

Kittens inherit features from both parents, but each kitten gets a different mix, so there is variation between them.

4 **Tanvir says, "If I grow big muscles at the gym, my children will be born with big muscles." Is he right? Explain.** [2]

No. Muscles built by exercise are not inherited; they are not passed on to children.

5 **Why must you never take medicine that belongs to someone else?** [1]

It may be the wrong medicine or the wrong amount for you and could make you ill.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A3 · LIFE CYCLES

What do seeds need to grow?

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Grade 4

Learning goals

- Plan a fair test.
- Present results in a bar chart.
- Draw conclusions.

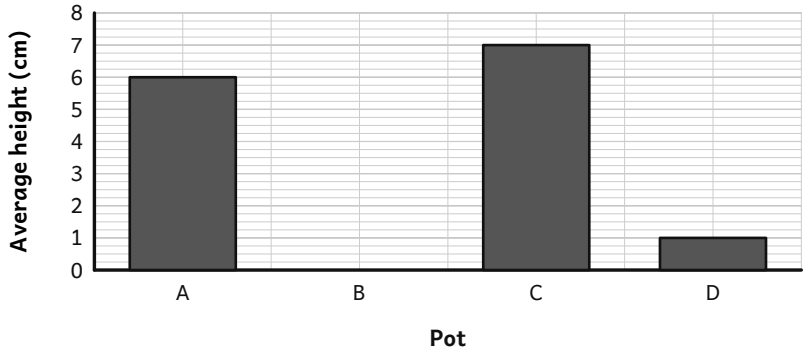
Class 4C planted 10 cress seeds in each of four pots. They changed one thing for each pot and measured the height of the seedlings after 7 days.

1 **Here is the plan. What did they change for each pot?** [2]

Pot	Water?	Light?	Warm?	Average height after 7 days (cm)
A	yes	yes	yes	6
B	no	yes	yes	0
C	yes	no (in a dark cupboard)	yes	7 (yellow and thin)
D	yes	yes	no (in a fridge)	1

Each pot was missing one thing: B no water, C no light, D no warmth. A had everything.

2 **Draw a bar chart of the heights.** [3]



3 **Pot C was tallest. Does that mean seedlings grow best in the dark? Explain.** [2]

No. They grew tall and thin looking for light, but were yellow and weak. Without light they cannot make food and will die.

4 **Write a conclusion.** [2]

Seeds need water and warmth to germinate; seedlings need light to grow into healthy green plants.

5 **Name two things they kept the same.** [1]

number of seeds, type of seed, size of pot, type of soil, amount of water

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A3 · LIFE CYCLES

Rajshahi silk: the life cycle of the silkworm

★★★
Grade 4

Learning goals

- Apply life cycle knowledge to a real industry.
- Use numbers from a real process.
- Think about the ethics of using animals.

Science · History · Economics · Ethics · Art

Rajshahi is famous for its beautiful **silk** saris. Silk is made by the **silkworm**, the caterpillar of a moth. Farmers grow **mulberry** trees, because silkworms eat only mulberry leaves.

A silk moth lays 300 to 500 tiny **eggs**. The silkworms (larvae) hatch and eat leaves day and night for about a month, growing about 10 000 times heavier! Then each one spins a **cocoon** around itself from a single thread of silk, which can be **one kilometre long**. Inside the cocoon it becomes a **pupa**. After about two weeks an adult **moth** comes out, and the cycle starts again.

To get long unbroken threads, most cocoons are put in hot water before the moth comes out, which kills the pupa. Some makers produce "peace silk" by waiting for the moths to leave first, but the thread is broken into shorter pieces.

Based on information about sericulture (silk farming) in Rajshahi, Bangladesh.

1 **Complete the silkworm life cycle.** [3]

- [5] adult moth lays eggs
- [1] egg
- [4] pupa inside the cocoon
- [2] silkworm (larva) eating mulberry leaves
- [3] spins a cocoon

2 **Why must silk farmers also be good at growing trees?** [1]

Silkworms eat only mulberry leaves, so farmers must grow lots of healthy mulberry trees.

3 **Maths. Suppose one sari needs 3000 cocoons. A farmer's silkworms make 12 000 cocoons. How many saris can be made?** [2]

12 000 ÷ 3000 = 4 saris

4 **What might each person think about killing the pupae to make silk?** [3]

Person	View
a silk farmer	It is our living and a tradition; we need long threads to make good silk.
an animal-welfare worker	Many insects die; peace silk is kinder to animals.
a sari buyer	I want beautiful silk, but I could choose peace silk if it is labelled.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A3 · LIFE CYCLES

Show what you know

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Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

R	F	L	S	U	E	U	D	W	D
W	L	A	D	U	L	T	K	G	I
U	S	I	N	H	E	R	I	T	P
I	E	L	Y	M	C	Y	C	L	E
H	E	T	A	D	P	O	L	E	U
U	D	E	E	G	G	F	M	J	Y
R	B	H	A	T	C	H	O	H	G
B	A	I	J	W	M	V	A	L	G
K	P	U	P	A	L	A	R	V	A
J	M	G	M	I	S	B	L	H	P

adult cycle egg hatch inherit larva pupa seed tadpole

2 Which animal goes through metamorphosis? [1]

- ○ chicken
- ○ cow
- ● **butterfly**
- ○ goat

3 What does a seed need to germinate? [1]

- ● **water and warmth**
- ○ light only
- ○ soil only
- ○ wind

4 Draw the life cycle of a plant you know, such as rice or mango. Label four stages. [4]

My plant life cycle

seed → seedling → adult plant with flowers → fruit with seeds (back to seed).

5 **Challenge. Many frogs in Bangladesh are disappearing because ponds and wetlands are being filled in for building. Use the frog life cycle to explain why losing ponds is a problem for frogs.** [2]

Frogs lay eggs in water and tadpoles can only live in water, so without ponds they cannot complete their life cycle and no young frogs grow up.